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MODEL-II (M/M/1):(N/FCFS) Numerical-03

Q: If for a period of 2 hours in the day (8-10)am trains arrive at the yard every 20 min but the service time continues to remain 36 minutes, then calculate for this period

(a) the probability that the yard is empty, and

(b) the average no. of train in the system, on the assumption that the line capacity of the yard is limited to 4 trains only.

Solution: $\lambda = \frac{1}{20}$ per min ; $N = 4$

$\mu = \frac{1}{36}$ per min

$$\rho = \frac{\lambda}{\mu} = \frac{\frac{1}{20}}{\frac{1}{36}} = \frac{36}{20} = 1.8$$

(i) $P_0 = \frac{1-\rho}{1-\rho^{N+1}}$

$= \frac{1-1.8}{1-(1.8)^5} = \frac{-0.8}{-17.8956} = 0.04$

(ii) Average no. of train in the system,

$L_s = P_0 \sum_{n=0}^{N} n\rho^n \quad (N=4)$

$= 0.04 \left[0 + \rho + 2\rho^2 + 3\rho^3 + 4\rho^4\right]$

$= (0.04)(0 + 1.8 + 6.48 + 17.496 + 41.9904)$

$= (0.04)(67.7664)$

$= 2.71 = 3 \text{ (approx)}$

$L_s = 3$ train.

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Model III (M/M/S) : (∞/FCFS) (Multiservice Model)

1) $$P_n = \begin{cases} \frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n P_0 & , n = 0, 1, 2 \cdots S-1 \\ \frac{1}{S!} \cdot \frac{1}{S^{n-S}} \left(\frac{\lambda}{\mu}\right)^n P_0 & , n = S, S+1 \cdots \end{cases}$$

2) $$P_0 = \left[\sum_{n=0}^{S-1} \frac{(SP)^n}{n!} + \frac{(SP)^S}{S!(1-P)} \right]^{-1}$$

3) Traffic intensity, $P = \frac{\lambda}{\mu S}$

4) Length of queue, $L_q = P_S \frac{P}{(1-P)^2}$, where $P_S = \frac{\left(\frac{\lambda}{\mu}\right)^S P_0}{S!}$

5) Length of the system, $L_s = L_q + \frac{\lambda}{\mu}$

6) Waiting time in the system, $W_s = \frac{L_s}{\lambda}$ or $W_q + \frac{1}{\mu}$

7) Waiting time in the queue, $W_q = \frac{L_q}{\lambda}$ or $P_S \cdot \frac{1}{S\mu(1-P)^2}$

8) The mean no. of individuals who actually wait is given by, $L(L>0) = \frac{1}{1-P}$

9) The mean waiting time in the queue for those who actually wait is given by $W(W>0)$,

$$= \frac{1}{s\mu - \lambda}$$

10) Probability $P(W>0) = \frac{P_s}{1-P}$

11) Probability that there will be someone waiting $= \frac{P_s P}{1-P}$

12) Average number of idle servers = S − (Avg. no. of customer served)

13) Efficiency of M/M/S model = $\frac{\text{Avg. no. of customers served}}{\text{Total no. of customers served.}}$

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Model III : (Multiservice Model) (M/M/S) : (∞/FCFS)

Numerical Problem - 01

Q : A supermarket has two girls ringing up sales at the counters. If the service time for each customer is exponential with mean four minutes and if people arrive in a Poisson fashion at the counter, at the rate of 10 per hour, then calculate,

(i) The probability of ~~having~~ customer to wait for service.

(ii) The expected percentage of idle time for each girl.

(iii) If a customer has to wait, find the expected length of his waiting time.

Solution: Probability of waiting time for service is,

$$P(W>0) = \frac{P_s}{1-\rho} = \frac{\left(\frac{\lambda}{\mu}\right)^s P_0}{s!\,(1-\rho)}$$

Given,

$$\lambda = \frac{10}{60} \text{ per min.} = \frac{1}{6} \text{ per min.}$$

$$\mu = \frac{1}{4} \text{ per min.}$$

$$s = 2$$

$$\rho = \frac{\lambda}{\mu s} = \frac{\frac{1}{6}3}{\frac{1}{4}\times 2} = \frac{1}{3}$$

We Know,

$$\therefore P_0 = \left[\sum_{n=0}^{s-1} \frac{(s\rho)^n}{n!} + \frac{(s\rho)^s}{s!\,(1-\rho)}\right]^{-1}$$

$$P_0 = \left[\sum_{n=0}^{1} \frac{\left(2 \cdot \frac{1}{3}\right)^n}{n!} + \frac{\left(2 \cdot \frac{1}{3}\right)^2}{2!\left(1-\frac{1}{3}\right)} \right]^{-1}$$

$0! = 1$

$a^0 = 1$

$$= \left[\left(1 + \frac{2}{3}\right) + \frac{\frac{4}{9}}{2 \times \frac{2}{3}} \right]^{-1}$$

$$= \left[\frac{5}{3} + \frac{1}{3} \right]^{-1} = \left(\frac{6}{3}\right)^{-1} = 2^{-1} = \frac{1}{2}$$

Thus, probability $(w>0) = \dfrac{\left(\frac{\lambda}{\mu}\right)^s P_0}{s!\left(1-\rho\right)} = \dfrac{\left(\frac{\frac{1}{3}}{\frac{1}{2}}\right)^2 \times \frac{1}{2}}{2!\left(1-\frac{1}{3}\right)}$

$$= \frac{\left(\frac{2}{3}\right)^2 \times \frac{1}{2}}{2 \times \frac{2}{3}} = \frac{\frac{4}{9} \times \frac{1}{2}}{2 \times \frac{2}{3}} = \frac{1}{6}$$

(ii) The fraction of the time the service remain busy, (i.e traffic intensity) is given by,

$$\rho = \frac{\lambda}{\mu S} = \frac{1}{3}$$

fraction of the time the service remains idle is,

$$= \left(1 - \frac{1}{3}\right) = \frac{2}{3}.$$

percentage of idle time of each girl $= \left(\frac{2}{3} \times 100\right)\%$

$= 67\%$.

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(iii) Expected length of customer who has to wait in queue,

$$(W/W>0) = \frac{1}{s\mu - \lambda}$$

$$= \frac{1}{2 \times \frac{1}{4_2} - \frac{1}{6}}$$

$$= \frac{1}{\frac{1}{2} - \frac{1}{6}} = \frac{1}{\frac{3-1}{6}}$$

$$= \frac{1}{\frac{2}{6_3}} = \frac{1}{\frac{1}{3}} = 3 \text{ min.}$$

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Model III : (Multiservice Model) (M/M/S): (∞/FCFS)

Numerical Problem - 02

Q: Four counters are being run on the frontier of a country to check the passports and necessary papers of the tourists. The tourists choose any counter at random If the arrival at the frontier is poisson at the rate λ and the service time is exponential with parameter $\frac{\lambda}{2}$, What is the steady state average queue at each counter?

Solution:

$S = 4$

$\lambda = \lambda$

$\mu = \frac{\lambda}{2}$

$\rho = \frac{1}{2}$

$\therefore \rho = \frac{\lambda}{\mu S}$

$\rho = \frac{\lambda}{\frac{\lambda}{2} \times 4^2} = \frac{1}{2}$

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Expected queue length, $L_q = P_s \frac{\rho}{(1-\rho)^2}$

$$= \frac{\left(\frac{\lambda}{\mu}\right)^s P_0}{s!} \cdot \frac{\rho}{(1-\rho)^2}$$

Thus,

$$P_0 = \left[\sum_{n=0}^{s-1} \frac{(s\rho)^n}{n!} + \frac{(s\rho)^s}{s!(1-\rho)} \right]^{-1}$$

$$= \left[\sum_{n=0}^{3} \frac{\left(4 \cdot \frac{1}{2}\right)^n}{n!} + \frac{\left(4 \cdot \frac{1}{2}\right)^4}{4!\left(1-\frac{1}{2}\right)} \right]^{-1}$$

$4! = 4 \times 3 \times 2 \times 1$

$2^0 = 1$

$0! = 1$

$$= \left[\sum_{n=0}^{3} \frac{2^n}{n!} + \frac{2^4}{24 \times \frac{1}{2}} \right]^{-1}$$

$$= \left[\left(1 + 2 + \frac{2^2}{2!} + \frac{2^3}{3!}\right) + \frac{16^4}{123} \right]^{-1}$$

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$$= \left[\left(1 + 2 + \frac{4^2}{2} + \frac{8^4}{63}\right) + \frac{4}{3} \right]^{-1}$$

$$= \left[5 + \frac{4}{3} + \frac{4}{3} \right]^{-1}$$

$$= \left[\frac{15+4+4}{3} \right]^{-1} = \left[\frac{23}{3} \right]^{-1} = \frac{3}{23}$$

$$\therefore P_0 = \frac{3}{23}$$

So, Expected queue length $= \frac{\left(\frac{\lambda}{\mu}\right)^5 P_0}{5!} \cdot \frac{\rho}{(1-\rho)^2}$

$$= \frac{\left(\frac{\lambda}{\lambda/2}\right)^4 \cdot \frac{3}{23}}{4!} \cdot \frac{\frac{1}{2}}{\left(1-\frac{1}{2}\right)^2}$$

$$= \frac{2^4 \cdot \frac{3}{23}}{24} \cdot \frac{\frac{1}{2}}{\frac{1}{4}}$$

$$= \frac{4}{\cancel{16}} \times \frac{3}{23} \times \frac{1}{\cancel{24}_{12}} \times \frac{1}{\cancel{2}} \times \cancel{4}^{2}$$

$$= \frac{4}{23}$$

$$\therefore L_q = \frac{4}{23}.$$

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Model IV (M/M/S) : (N/FCFS)

This model is essentially the same as model III, except that the maximum number of customers in the system is limited to N, where $N > S$ (S = No. of channels)

$$\therefore \quad \lambda_n = \begin{cases} \lambda, & 0 \le n \le N \\ 0, & n \ge N \end{cases}$$

$$\mu_n = \begin{cases} n\mu, & 0 \le n \le S \\ S\mu, & S \le n \le N \end{cases}$$

Formula:

1) $$P_n = \begin{cases} \frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n P_0, & 0 \le n \le S \\ \frac{1}{S^{n-S}\, S!}\left(\frac{\lambda}{\mu}\right)^n P_0, & S \le n \le N \end{cases}$$

2) $$P_0 = \left[\sum_{n=0}^{S-1} \frac{1}{n!} \left(\frac{\lambda}{\mu}\right)^n + \sum_{n=S}^{N} \frac{1}{S^{n-S} S!} \left(\frac{\lambda}{\mu}\right)^n \right]^{-1}$$

3) $$L_q = \sum_{n=S}^{N} (n-S) P_n$$

4) $$L_s = L_q + S - P_0 \sum_{n=0}^{S-1} \frac{(S-n)(Sf)^n}{n!}$$

5) $$W_q = W_s - \frac{1}{\mu}$$

6) $$W_s = \frac{L_s}{\lambda(1-P_N)}$$

7) Traffic intensity, $f = \frac{\lambda}{\mu S}$ or, $Sf = \frac{\lambda}{\mu}$

Numerical Problems on (M/M/S) : (N/FCFS)

Q: A barber shop has two barber and three chairs for customers. Assume that the customers arrive in a poisson fashion at a rate of five per hour and each barber services customers according to an exponential distribution with 15 minutes. Further, if a customer arrives and there are no empty chairs in the shop, he will leave. What is the expected no. of customers in the shop?

Solution:

$S = 2$

$N = 3$

$\lambda = \frac{5}{60} = \frac{1}{12}$ per min

$\mu = \frac{1}{15}$ per min,

$\frac{\lambda}{\mu} = \frac{1}{12} \times 15 = \frac{5}{4}$

$\rho = \frac{\lambda}{\mu S}$

$S\rho = \frac{\lambda}{\mu} = \frac{5}{4}$

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$$P_0 = \left[\sum_{n=0}^{S-1} \frac{1}{n!} \left(\frac{\lambda}{\mu}\right)^n + \sum_{n=S}^{N} \frac{1}{S^{n-S}\, S!} \left(\frac{\lambda}{\mu}\right)^n \right]^{-1}$$

$$= \left[\sum_{n=0}^{1} \frac{1}{n!} \left(\frac{5}{4}\right)^n + \sum_{n=2}^{3} \frac{1}{2^{n-2}\, 2!} \left(\frac{5}{4}\right)^n \right]^{-1}$$

$0! = 1$

$a^0 = 1$

$$= \left[\left(1 + \frac{5}{4}\right) + \left(\frac{1}{2} \times \frac{25}{16} + \frac{1}{2 \times 2} \left(\frac{125}{64}\right)\right) \right]^{-1}$$

$$= \left[1 + \frac{5}{4} + \frac{25}{32} + \frac{125}{256} \right]^{-1}$$

$$= \left[\frac{256 + 320 + 200 + 125}{256} \right]^{-1}$$

$$= \left[\frac{901}{256} \right]^{-1} = \frac{256}{901} = 0.28$$

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$$P_n = \begin{cases} \frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n P_0 , & 0 \le n < s \\ \frac{1}{s^{n-s} s!}\left(\frac{\lambda}{\mu}\right)^n P_0 , & s \le n \le N \end{cases}$$

if $n=3$,

$0 \le 3 < 2$ ×

$2 \le 3 \le 3$ ✓

$$= \begin{cases} \frac{1}{n!}\left(\frac{5}{4}\right)^n \times 0.28 , & 0 \le n < s \\ \frac{1}{2^{n-2} \times 2!}\left(\frac{5}{4}\right)^n \times 0.28 , & s \le n \le N \end{cases}$$

Thus, $L_s = L_q + s - P_0 \sum_{n=0}^{s-1} \frac{(s-n)(sf)^n}{n!}$

$$= \sum_{n=s}^{N} (n-s) P_n + s - P_0 \sum_{n=0}^{s-1} \frac{(s-n)(sf)^n}{n!}$$

$$= \sum_{n=2}^{3} (n-2)P_n + 2 - 0.28 \sum_{n=0}^{1} \frac{(2-n)\left(\frac{5}{4}\right)^n}{n!}$$

$$= 0 + P_3 + 2 - 0.28\left(2 + 1.25\right)$$

$$= P_3 + 2 - 0.28 \times 3.25$$

$$= P_3 + 1.09$$

$$= \left[\frac{1}{2^{3-2} \times 2!}\left(\frac{5}{4}\right)^3 \times 0.28\right] + 1.09$$

$$= \left[\frac{1}{4} \times (1.25)^3 \times 0.28\right] + 1.09$$

$$= \frac{1.9531 \times 0.28}{4} + 1.09 \qquad = \frac{0.5468}{4} + 1.09$$

$$= 1.226$$